EV Battery Course — Module 14: Battery Recycling and Circular Economy

Oct 9, 2026 · @Ashishpreet

Recycling economics are driven by what is inside the battery: nickel- and cobalt-rich packs can pay for their own collection and processing; LFP packs mostly cannot without a gate fee or policy support. This module follows a pack from collection to recovered metals, compares pyrometallurgy, hydrometallurgy and direct recycling, covers India's Battery Waste Management Rules, and works a hypothetical business case.

1. Learning objectives

- Describe the end-of-life value chain: collection, identification, SOH triage, discharge, dismantling, reuse, shredding, black mass and metal recovery.
- Compare pyrometallurgy, hydrometallurgy and direct recycling on cost, recovery, energy, environment and maturity.
- Explain battery passports, EPR and India's rules.
- Build and test a recycling business case with stated assumptions.

2. Fundamentals: the waste hierarchy for batteries

1. **Reuse** in the same application (repair, module replacement).
2. **Repurpose** (second life, Module 10).
3. **Recycle** materials.
4. **Dispose** (not permitted for EV batteries in India under the BWMR).

3. The value chain

```
flowchart LR
  A["Collection and reverse logistics"] --> B["Identification and chemistry
sorting"] --> C["SOH testing and triage"]
  C -- "good" --> D["Reuse / second life"]
  C -- "end of life" --> E["Safe discharge"] --> F["Pack and module
dismantling"] --> G["Shredding and separation"] --> H["Black mass"] -->
I["Pyro / hydro / direct recycling"] --> J["Battery-grade materials"]
  G --> K["Copper, aluminium, steel, plastics"]
```

Stage	Key points
Collection and reverse logistics	Dangerous-goods rules; damaged packs need special containers; high cost per km for heavy packs; dealer and service networks are collection points
Identification	Labels, QR codes or battery passports give chemistry, manufacturer and history; unknown chemistries need analysis (XRF)
SOH assessment	Capacity and resistance test or BMS data; decides reuse vs recycle
Safe discharge	Resistive/electronic discharge (energy can be recovered) or salt-water immersion (slow, creates wastewater); prevents shorts during shredding
Dismantling	Mostly manual today; glued CTP packs are hard; robotic and automated disassembly in development
Shredding and separation	Shredding under inert atmosphere or liquid; sieving, magnetic, eddy-current and density separation yield copper, aluminium, steel, plastics and **black mass** (cathode + anode powders)

4. Recycling routes

Pyrometallurgy. Smelt cells or black mass at about 1,400–1,600 °C. Nickel, cobalt and copper form an alloy; lithium, aluminium and manganese end up in slag (lithium recovery possible but costly). Electrolyte and graphite burn as fuel.

Hydrometallurgy. Leach black mass in acid (often sulphuric acid with hydrogen peroxide); separate metals by solvent extraction and precipitation into nickel sulphate, cobalt sulphate, manganese sulphate and lithium carbonate or hydroxide. High recoveries, including lithium.

Direct recycling. Recover cathode material as a compound, then relithiate and restore it without breaking it down to elements. Potentially lowest energy and cost, especially valuable for LFP, but needs clean, sorted feed of known chemistry; mostly pilot stage.

Criterion	Pyrometallurgy	Hydrometallurgy	Direct recycling
Capital cost	High (furnaces)	Medium-high	Low-medium (projected)
Operating cost	Medium-high (energy)	Medium (chemicals)	Low (projected)
Ni, Co recovery	High (~95%)	High (~95–98%)	High as cathode material
Li recovery	Low unless slag is treated	High (~80–95%)	High
Graphite	Burned	Can be recovered (quality challenge)	Can be recovered
Energy use	High	Medium	Low
Environmental issues	CO_2, off-gas treatment	Wastewater, chemical reagents	Lowest (projected)
Feed flexibility	Very high (mixed feed)	High	Low (needs sorted chemistry)
Maturity	Commercial (e.g. Umicore)	Commercial and dominant in China; growing in India	Pilot

Many modern plants combine mechanical pre-treatment with hydrometallurgy.

5. Step-by-step: from end-of-life pack to new cathode

1. Pack arrives; passport/label read; BMS data checked if accessible.
2. SOH test: high SOH → second life; low SOH → recycling.
3. Discharged to near 0 V.
4. Dismantled: housing, busbars, BMS and cooling plates removed and sent to metal and electronics recyclers.
5. Modules or cells shredded; electrolyte evaporated and captured; black mass separated.
6. Black mass leached; impurities removed; Ni, Co, Mn and Li separated.
7. Metal salts sold to precursor and cathode makers; new cathode made; cell made: the loop closes.

6. Engineering equations

Material recovered:

```
m_{recovered} = m_{input} \times y_{black\,mass} \times w_{element} \times \eta_{recovery}
```

Lithium carbonate equivalent (LCE):

```
LCE = m_{Li} \times 5.323
```

Revenue, EBITDA and payback:

```
R = \sum_k m_k \times P_k \times f_{payable} \qquad EBITDA = R - C_{feed} - C_{opex} \qquad \text{Payback} \approx \frac{Capex}{EBITDA}
```

y = black-mass yield (fraction of cell mass), w = element mass fraction in black mass, η = recovery efficiency, P = price per tonne, f_payable = fraction of metal value paid for an intermediate product.

7. Hypothetical business case: 10,000 t/year hydrometallurgical plant

All figures are illustrative assumptions, not market quotes or a real project.

Assumption	NMC-rich feed
Input (cells and modules after dismantling)	10,000 t/year
Black-mass yield	40% → 4,000 t
Black-mass composition	Ni 22%, Co 3%, Li 3.5%
Recovery	Ni 95%, Co 95%, Li 85%, Cu 90%, Al 85%
Cu and Al in cells	8% and 5% of cell mass
Metal prices	Ni $16,000/t; Co $30,000/t; LCE $10,000/t; Cu $9,500/t (sold at 90%); Al $1,800/t
Payable for Ni/Co/Li salts	80% of contained metal value
Feed cost (paid to battery owner, net of EPR credits)	$1,000/t
Operating cost (chemicals, energy, labour, waste treatment)	$1,200/t of input

Assumption	NMC-rich feed
Capex	$60 million, 10-year depreciation

Results (NMC-rich feed)

Product	Recovered	Revenue
Nickel	836 t	$10.70 M
Cobalt	114 t	$2.74 M
Lithium	119 t Li = 633 t LCE	$5.07 M
Copper	720 t	$6.16 M
Aluminium	425 t	$0.77 M
Total revenue	—	**$25.4 M**

- Costs: feed $10.0 M + opex $12.0 M = $22.0 M → **EBITDA $3.4 M**.
- Depreciation $6.0 M → **EBIT −$2.6 M**; simple payback 60 ÷ 3.4 ≈ **17.5 years**: not attractive.
- **Sensitivity:** if feed cost falls to $500/t, EBITDA = $8.4 M and payback ≈ 7 years. Feed price and metal prices dominate the outcome.

LFP feed (same plant, Li 4.5% in black mass, no Ni/Co): Li revenue ≈ $6.5 M + Cu/Al $6.9 M = $13.4 M, against opex $12.0 M → only $1.4 M before paying anything for feed or depreciation. LFP recycling needs a **gate fee** (the battery owner pays the recycler), EPR-certificate revenue, subsidy, or lower-cost processes such as direct recycling.

8. India: rules, EPR and policy (as of October 2026)

Battery Waste Management Rules (BWMR), 2022, administered by the Central Pollution Control Board (CPCB):

- Producers (manufacturers and importers, including OEMs) must register and meet **extended producer responsibility (EPR)** targets through collection, refurbishment and recycling, using tradable EPR certificates from registered recyclers and refurbishers.
- Collected batteries may not go to landfill or incineration.
- **Material recovery targets for EV batteries** (share of dry weight): 70% in 2024–25, rising to 90% from 2026–27 onward.
- **EV battery collection target:** 70% of batteries placed on the market by 2027–28.

- **Domestic recycled content in new batteries:** at least 5% from 2027–28, rising to 20% by 2030–31.
- Several amendments since 2022 adjusted procedures (e.g. EPR portal and certificate rules); check the current CPCB text.

Critical mineral recycling incentive scheme: approved by the Union Cabinet on 3 September 2025 with ₹1,500 crore for FY2025–26 to FY2030–31. It covers e-waste, Li-ion battery scrap and other scrap; offers a 20% capex subsidy on plant and machinery plus an opex subsidy tied to incremental sales; capped at ₹50 crore per large entity; and targets at least 270,000 t/year of recycling capacity, yielding about 40,000 t/year of critical minerals.

Market reality: India's recycling industry today processes mainly consumer electronics and two-/three-wheeler batteries, plus manufacturing scrap. Large volumes of end-of-life EV car packs arrive only in the 2030s. Leading players include Attero, Lohum, Recyclekaro, BatX and others (capacities change quickly; check current data).

Battery passports and traceability: The EU Battery Regulation requires a digital battery passport for EV and industrial batteries from February 2027, including carbon footprint, materials, recycled content and SOH information. Indian exporters to the EU will need to comply; India has discussed similar digital tracking under BWMR.

9. Real-world examples

- **Umicore (Belgium):** pyro-hydro hybrid, long commercial history.
- **Brunp (CATL subsidiary), GEM (China):** large hydrometallurgical capacity.
- **Redwood Materials (US):** recycling plus cathode and anode materials production; also second-life storage.
- **Li-Cycle (US/Canada):** "spoke and hub" model; ran into financial difficulties in 2024–25 and was acquired by Glencore, illustrating how hard the economics are.
- **India:** Attero, Lohum and others operate hydrometallurgical recycling, with expansion plans tied to the new incentive scheme.

10. Manufacturing applications (design for recycling)

- Use labels and passports with chemistry information.
- Prefer fasteners or debondable adhesives over permanent glue where feasible.
- Standardise module and cell formats to enable automated dismantling.
- Keep production scrap (electrode trim, rejected cells) in a closed loop with recyclers: today this is the largest and cleanest feedstock.

- Plan EPR obligations in the costing: EPR certificate cost per kg is a real line item for OEMs in India.

11. Common mistakes and misconceptions

- **"Recycling is always profitable."** It depends on chemistry, metal prices and feed costs.
- **"Recycling can supply most demand soon."** Battery demand grows faster than end-of-life volumes; recycling covers a small share until the 2030s–2040s.
- **"95% recycling rate" is a single number.** Recovery differs by element; lithium and graphite are often lower.
- **"Discharging is optional."** Shredding charged batteries risks fires.

12. Practical case study: a fire at a recycling yard

Problem. A recycler stores incoming scooter packs in open bins; one damaged pack ignites and spreads.

Root causes. No triage of damaged packs; storage density too high; no thermal monitoring; packs not discharged.

Corrective action. Receiving inspection with quarantine for damaged or swollen packs; storage in fire-rated containers with spacing; thermal cameras; discharge before storage; emergency drills.

Lesson. The riskiest moment in recycling is often before processing starts.

13. Revision notes

- Reuse → repurpose → recycle.
- Black mass is the valuable intermediate.
- Pyro: robust, poor Li recovery, high energy. Hydro: high recoveries including Li. Direct: promising, pilot stage.
- NMC recycling can pay; LFP usually needs gate fees or support.
- India BWMR: EV recovery 70% (2024–25) → 90% (2026–27+); recycled content 5% (2027–28) → 20% (2030–31).

14. Glossary

Term	Meaning
Black mass	Shredded electrode powder concentrate
EPR	Extended producer responsibility
LCE	Lithium carbonate equivalent
Pyro / hydrometallurgy	Smelting / aqueous chemical processing
Direct recycling	Restoring cathode material without full breakdown
Gate fee	Fee paid to the recycler to accept waste
Battery passport	Digital record of a battery's materials and history

15. Knowledge check

Conceptual

1. Why is LFP less profitable to recycle than NMC?
2. Why does pyrometallurgy lose lithium?
3. Why must packs be discharged before shredding?
4. How does EPR create revenue for recyclers?
5. Why does direct recycling need sorted feed?

Numerical

1. 1,000 t NMC cells, 40% black mass, 22% Ni, 95% recovery. How much nickel is recovered?
2. Convert 50 t of recovered lithium to LCE.
3. A plant has EBITDA of $5 M and capex of $45 M. Simple payback?

Application

1. An Indian OEM must meet EPR targets but end-of-life volumes are tiny. What options exist?
2. Evaluate whether a recycler should build a pyrometallurgical or hydrometallurgical plant in India in 2027.

16. Answer key

Conceptual

1. LFP contains no nickel or cobalt; iron and phosphate are cheap; lithium content alone rarely covers processing costs.
2. Lithium oxidises and reports to the slag; extracting it from slag is costly.
3. Stored energy can cause sparks and fires during shredding.
4. Producers need EPR certificates; recyclers earn them by processing batteries and sell them to producers.
5. Its products are specific cathode compounds; mixed chemistries contaminate the product.

Numerical

1. 1,000 × 0.40 × 0.22 × 0.95 = **83.6 t**.
2. 50 × 5.323 = **266 t LCE**.
3. 45 ÷ 5 = **9 years**.

Application

1. Buy EPR certificates from registered recyclers handling other battery waste or production scrap where rules permit; partner with recyclers on take-back programmes; collect through dealers; invest in recycling or refurbishment partners; track obligations in costing.
2. Hydrometallurgy fits better: Indian feed is increasingly LFP and mixed, lithium recovery is needed for the targets, capex can scale modularly, and the 2025 incentive supports it; pyrometallurgy suits large, Ni/Co-rich, mixed feed and needs high volumes, which India lacks today.

17. Further learning

- Harper et al., "Recycling lithium-ion batteries from electric vehicles," *Nature* 2019.
- IEA *Global Critical Minerals Outlook 2025* (recycling chapter).
- CPCB Battery Waste Management Rules 2022 and amendments; EPR portal.
- EU Battery Regulation (EU) 2023/1542.

Sources

- IEA policy database: Battery Waste Management Rules 2022
- PIB/PM India: Cabinet approves ₹1,500 crore critical mineral recycling scheme (3 Sep 2025)

Next: Module 15, the global and Indian battery industry.